

● HARD

● GEOMETRY AND TRIGONOMETRY

Lines, angles, and triangles

01

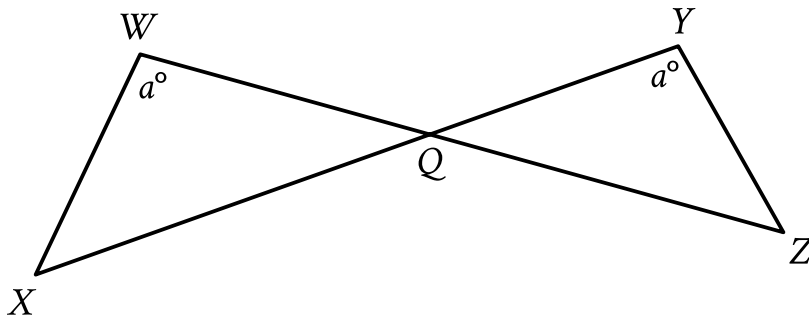
10 questions · ~15 min

Q1

GEOMETRY AND TRIGONOMETRY

Triangles PQR and LMN are graphed in the xy -plane. Triangle PQR has vertices P , Q , and R at $4, 5$, $4, 7$, and $6, 5$, respectively. Triangle LMN has vertices L , M , and N at $4, 5$, $4, 7 + k$, and $6 + k, 5$, respectively, where k is a positive constant. If the measure of $\angle Q$ is t° , what is the measure of $\angle N$?

- A. $90 - t - k^\circ$
- B. $90 - t + k^\circ$
- C. $90 - t^\circ$
- D. $90 + k^\circ$

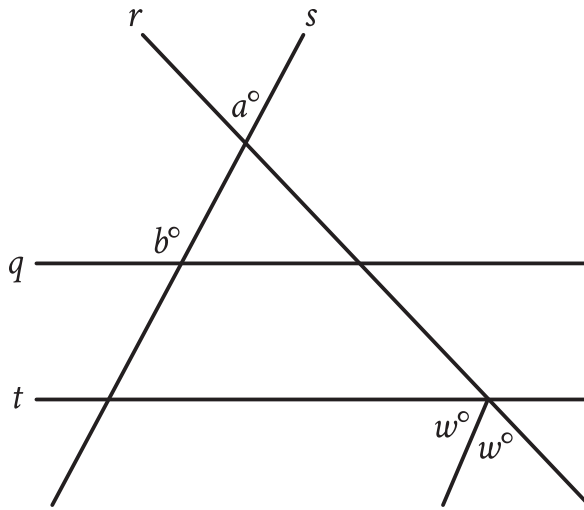


Note: Figure not drawn to scale.

- Triangle WXQ has a common vertex with triangle YZQ .
- The common vertex is Q .
- For triangle WXQ :
 - The measure of angle W is a° .
- For triangle YZQ :
 - The measure of angle Y is a° .
- A note indicates the figure is not drawn to scale.

In the figure shown, $\overline{WZ}$ and $\overline{XY}$ intersect at point Q , $YQ = 21$, $WQ = 70$, $WX = 60$, and $XQ = 120$. What is the length of $\overline{YZ}$?

- A. 18
- B. 36
- C. 120
- D. 200



Note: Figure not drawn to scale.

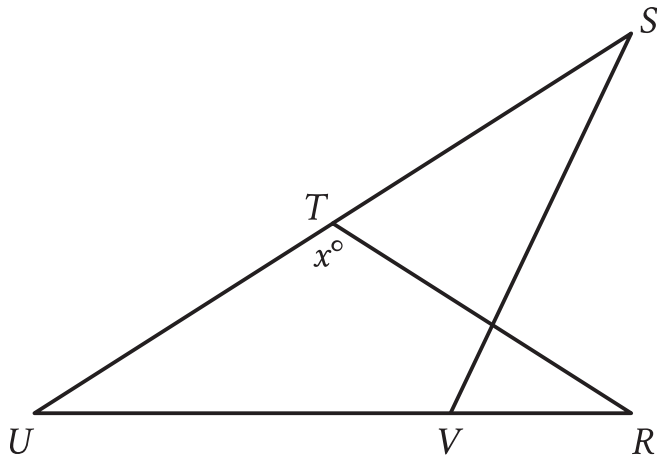
- Counterclockwise from top right, the lines are labeled s , r , q , and t .
- Line s intersects line r above line q .
- Line s and line r each intersect line q and line t .
- At the intersection of line s and line r , 1 angle is labeled counterclockwise from top as follows:
 - Top: a°
- At the intersection of line s and line q , 1 angle is labeled counterclockwise from top-left as follows:
 - Top left: b°
- At the intersection of line r and line t :
 - The line segment divides the bottom-left angle into two equal angles, each labeled w° .
- A note indicates the figure is not drawn to scale.

In the figure, parallel lines q and t are intersected by lines r and s . If $a = 43$ and $b = 122$, what is the value of w ?

STUDENT-PRODUCED RESPONSE

	$\frac{\quad}{\quad}$	$\frac{\quad}{\quad}$	$\frac{\quad}{\quad}$
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



Note: Figure not drawn to scale.

- Triangle $\triangle RST$ is labeled as follows:
 - The measure of angle $\angle RST$ is x° .
- The 2 triangles partially overlap:
 - Side RS of triangle $\triangle RST$ intersects side SV of triangle $\triangle TUV$.
 - Vertex V of triangle $\triangle TUV$ lies on side RU of triangle $\triangle RST$.
 - Vertex T of triangle $\triangle RST$ lies on side SV of triangle $\triangle TUV$.
- A note indicates the figure is not drawn to scale.

In the figure, $RT = TU$, the measure of angle $\angle VST$ is 29° , and the measure of angle $\angle RVS$ is 41° . What is the value of x ?

STUDENT-PRODUCED RESPONSE

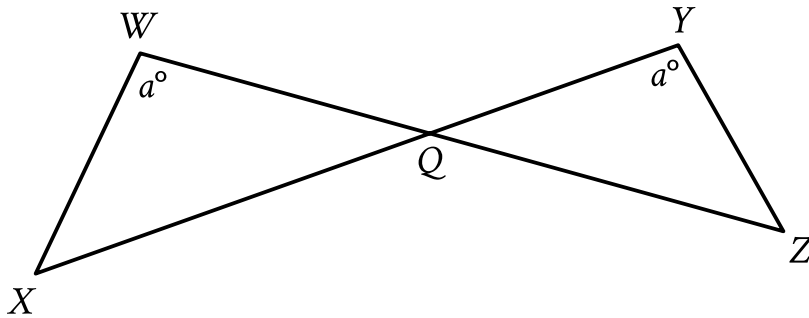
.	/	.	.
0	0	0	0

1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Quadrilaterals $PQRS$ and $WXYZ$ are similar, where P , Q , and R correspond to W , X , and Y , respectively. The measure of $\angle S$ is 135° , $PS = 45$, and $WZ = 9$. What is the measure of $\angle Z$?

- A. 5°
- B. 27°
- C. 45°
- D. 135°



Note: Figure not drawn to scale.

- The 2 triangles intersect at vertex upper Q.
- The measure of angle upper X upper W upper Q is a° .
- The measure of angle upper Z upper Y upper Q is a° .
- A note indicates the figure is not drawn to scale.

In the figure shown, $\overline{WZ}$ and $\overline{XY}$ intersect at point Q. $YQ = 63$, $WQ = 70$, $WX = 60$, and $XQ = 120$. What is the length of $\overline{YZ}$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Triangle XYZ is similar to triangle RST such that X , Y , and Z correspond to R , S , and T , respectively. The measure of $\angle Z$ is 20° and $2XY = RS$. What is the measure of $\angle T$?

- A. 2°
- B. 10°
- C. 20°
- D. 40°

Q8

GRID-IN

GEOMETRY AND TRIGONOMETRY

In triangle XYZ , angle Y is a right angle, point P lies on $\overline{XZ}$, and point Q lies on $\overline{YZ}$ such that $\overline{PQ}$ is parallel to $\overline{XY}$. If the measure of angle XZY is 63° , what is the measure, in degrees, of angle XPQ ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

GRID-IN

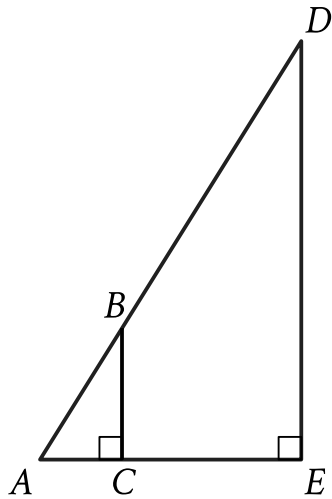
GEOMETRY AND TRIGONOMETRY

In triangle RST , angle T is a right angle, point L lies on $\overline{RS}$, point K lies on $\overline{ST}$, and $\overline{LK}$ is parallel to $\overline{RT}$. If the length of $\overline{RT}$ is 72 units, the length of $\overline{LK}$ is 24 units, and the area of triangle RST is 792 square units, what is the length of $\overline{KT}$, in units?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



Note: Figure not drawn to scale.

- Angle upper D upper E upper A is a right angle.
- Point upper B lies on side upper A upper D.
- Point upper C lies on side upper A upper E.
- Line segment upper B upper C is drawn from side upper A upper D to side upper A upper E to form right triangle upper A upper B upper C.
- Angle upper B upper C upper A is a right angle.
- A note indicates the figure is not drawn to scale.

In the figure shown, $AB = \sqrt{34}$ units, $AC = 3$ units, and $CE = 21$ units. What is the area, in square units, of triangle ADE ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
(0)	(0)	(0)	(0)
(1)	(1)	(1)	(1)
(2)	(2)	(2)	(2)
(3)	(3)	(3)	(3)
(4)	(4)	(4)	(4)
(5)	(5)	(5)	(5)

6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.